CASE 2 Edge case beam (heavy reinforcement + axial load) according to ACI 318-14

b x h = 30 x 60, cover = 3 cm, f_c' = 350 ksc, f_y = 4000 ksc, f_{yt} = 4000 ksc, stirrup DB10@70

Section	Top rebar	Bottom rebar	Side rebar	M_u (kg-m)	V_u (kg)	T_u (kg-m)	P_u (kg)
INITIAL	4-DB25 4-DB25	2-DB12	-	−54,000	30,000	300	-5,000
MID	2-DB12	2-DB20	-	11,000	5,000	0	12,000
END	3-DB25	3-DB16	4-DB16	-22,000	45,000	1,500	6,000

Test setup: High-strength concrete ($f'_c = 350$ ksc), requiring interpolated $\beta_1 = 0.80$ (not 0.85). INITIAL: Two-layer top reinforcement (8-DB25) with axial tension ($P_u = -5{,}000$ kg), doubly reinforced with yielded compression steel (2-DB12). Net tensile strain falls in transition zone if evaluated at centroid d ($0.004 \le \epsilon_t < 0.005, \phi = 0.883$) vs tension-controlled at d_t ($\epsilon_t \ge 0.005, \phi = 0.900$). Axial tension reduces V_c and T_{th} (torsion neglected per $T_u \le T_{th}$). MID: A_s (2-DB20) safely exceeds $A_{s,\min}$, compression steel (2-DB12) is elastic, and axial compression ($P_u = +12{,}000$ kg) satisfies beam limit ($P_u \le 0.10 f'_c A_g$) while enhancing V_c and T_{th}. END: T_u exceeds T_{th}, requiring full torsion design (A_t/s, A_l with 4-DB16 side bars, combined shear-torsion web crushing check). With $V_s > 1.10\sqrt{f'_c}bd$, stirrup spacing is governed by $s_{\max} = \min(d/4, 30\text{ cm})$, with axial compression $P_u = +6{,}000$ kg.

1. Design Parameters and material properties

1.1. Cross-section geometry

Beam width (b) = 30 cm

Total depth (h) = 60 cm

Gross area (A_g) = b x h = 30 x 60 = 1,800 cm^2

Concrete cover = 3 cm

1.2. Material Specifications

Concrete compressive strength (f_c') = 350 ksc

Main steel yield strength (f_y) = 4000 ksc (SD40)

Stirrup steel yield strength (f_{yt}) = 4000 ksc

Elasticity of steel (Es) = 2.04 x 10^6 ksc

Stress block factor

$$\beta_1 = 0.85 - \frac{0.05 \cdot (f'_c - 280)}{70} = 0.85 - \frac{0.05 \cdot (350 - 280)}{70} = 0.800$$

1.3. Strength reduction factors from table 21.2.1

Flexure = 0.65 to 0.90 (Determined by εt per Table 21.2.1)

Shear = 0.85

Torsion = 0.75

2. Critical section initial (support negative moment)

2.1 Geometry and factored demand

M_u = -54,000 kg-m = -54 t-m

V_u = 30,000 kg = 30 tons

T_u = 300 kg-m = 0.3 t-m

P_u = -5,000 kg = -5 tons

Top layer 1 4-DB25 (A_s = 4 x 4.909 = 19.64 cm^2)

Top layer 2 4-DB25 (A_s = 4 x 4.91 = 19.64 cm^2) and clearing space is 2.50 cm

Bottom 2-DB12 (A'_s = 2 x 1.131 = 2.262 cm^2)

Effective depth (d) of 2 Layer

$$y_1 = c_{\text{cov}} + d_{st} + \frac{d_b}{2} = 3.00 + 1.00 + \frac{2.50}{2} = 5.25 \text{ cm}$$

$$d_t = h - y_1 = 60.00 - 5.25 = 54.75 \text{ cm}$$

$$y_2 = y_1 + d_b + s_{\text{clear}} = 5.25 + 2.50 + 2.50 = 10.25 \text{ cm}$$

$$y_{\text{bar}} = \frac{(4 \times 5.25) + (4 \times 10.25)}{8} = 7.75 \text{ cm}$$

$$d_1 = h - y_{\text{bar}} = 60.00 - 7.75 = 52.25 \text{ cm}$$

Compression steel depth from bottom face (d_2)

$$d' = c_{\text{cov}} + d_{st} + \frac{d_{b,12}}{2} = 3.00 + 1.00 + \frac{1.20}{2} = 4.60 \text{ cm}$$

Rebar clearance and spacing limits (ACI 25.2.1)

$$s_{\text{clear}} = \frac{b - 2c_{\text{cov}} - 2d_{st} - n \cdot d_b}{n - 1} = \frac{30 - (2 \times 3.0) - (2 \times 1) - (4 \times 2.5)}{4 - 1} = 4 \text{ cm}$$

$$s_{\text{clear,min}} = \max(d_b,\ 2.5 \text{ cm}) = \max(2.5,\ 2.5) = 2.50 \text{ cm}$$

$s_{\text{clear}}(4.0 \text{ cm}) \geq s_{\text{clear,min}}(2.50 \text{ cm})$ PASS

2.2. Flexural strength analysis

Force equilibrium and neutral axis depth

$$C_c + C'_s - T_s = P_u \ (where\ P_u\ =\ -5{,}000\ kg, axial\ tension)$$

$$C_c + C'_s = T_s + P_u\ = T_s - |\ P_u\ |$$

Check yielding of tension steel T_s

Assume all layers of tension steel yield ($f_{s1} = f_{s2} = f_y = 4{,}000$)

$$T_s = A_s \cdot f_y = 39.270 \times 4000 = 157{,}080.0 \text{ kg}$$

Check yielding of compression steel (C'_s)

The stress in compression steel is $f'_s = \min(f_y,\ E_s\epsilon'_s)$ accounting for the displaced concrete area ($0.85f'_c = 297.5$ ksc)

Assume compression steel yields ($f'_s = f_y = 4{,}000$ ksc) accounting for displaced concrete area ($0.85f'_c = 297.5$ ksc)

$$C'_s = A'_s \cdot (f'_s - 0.85f'_c)$$

$$C'_s = 2.262 \times (4000 - 297.5) = 8{,}375.1 \text{ kg}$$

$$C_c = 0.85f'_c\beta_1 cb = 0.85 \times 350 \times 0.80 \times c \times 30.0 = 7{,}140.0 \cdot c$$

Substitute values for

$$7140.0 \cdot c + A'_s(f'_s - 297.5) = 157{,}080.0 - 5{,}000.0 = 152{,}080.0 \text{ kg}$$

$$7140.0 \cdot c + 8{,}375.1 = 152{,}080.0$$

$$c = \frac{143{,}704.9}{7140.0} = 20.127 \text{ cm}$$

$$\epsilon'_s = 0.003 \cdot \left(\frac{c - d'}{c}\right) = 0.003 \cdot \left(\frac{20.127 - 4.60}{20.127}\right) = 0.002315$$

$$\epsilon_y = \frac{f_y}{E_s} = \frac{4000}{2{,}040{,}000} = 0.001961$$

$$\text{Since } \epsilon'_s\ (0.002315) \geq \epsilon_y\ (0.001961)$$

$$f'_s = f_y = 4{,}000 \text{ ksc} \quad \text{(YIELDED CONFIRMED)}$$

Check the depth of the compression block (a) compared to (d')

$$a = \beta_1 \cdot c = 0.80 \times 20.127 = 16.102 \text{ cm}$$

$$a\ (16.102 \text{ cm}) > d'\ (4.60 \text{ cm})$$

Tension Strain Check (ϵ_t) and Strength Reduction Factor (φ)

$$\epsilon_{t,\text{centroid}} = 0.003 \cdot \left(\frac{d - c}{c}\right) = 0.003 \cdot \left(\frac{52.25 - 20.127}{20.127}\right) = 0.004788$$

$$\epsilon_{t,\text{outer}} = 0.003 \cdot \left(\frac{d_t - c}{c}\right) = 0.003 \cdot \left(\frac{54.75 - 20.127}{20.127}\right) = 0.005161$$

$$\epsilon_{t,\text{outer}} = 0.005161 \geq 0.00500$$

$$\phi = 0.900 \text{ (Tension} - \text{Controlled)}$$

According to ACI 318-14 Section 21.2.2, the determination of the section's behavior and the strength reduction factor (ϕ) must be based exclusively on the net tensile strain (ϵ_t) at the extreme tension steel layer (d_t).

Note: If an engineer incorrectly uses the strain at the steel centroid (ϵt,centroid = 0.004788), the cross-section would be misinterpreted as falling within the Transition Zone (0.00196 <ϵt<0.005). This would penalize the design by immediately reducing the ϕ factor to 0.883

$$A_{s,\min} = \max\left(\frac{14}{f_y}, \frac{0.8\sqrt{f'_c}}{f_y}\right) \cdot b \cdot d = \max\left(\frac{14}{4000}, \frac{0.8\sqrt{350}}{4000}\right) \cdot 30 \cdot 52.25 = 5.866 \text{ cm}^2$$

Check with actual reinforcement (8-DB25 = 39.27 cm^2):

$$A_{s,prov}(39.270\ cm^2) \geq A_{s,min}(5.866\ cm^2) \text{ PASS}$$

Moment from concrete compression (M_{Cc})

$$C_c = 0.85 f'_c \beta_1 c b = 0.85 \times 350 \times 0.80 \times 20.127 \times 30.0 = 143{,}706.78 \text{ kg}$$

$$\text{Arm} = \frac{h}{2} - \frac{a}{2} = 30.00 - \frac{16.102}{2} = 21.949 \text{ cm}$$

$$M_{Cc} = 143{,}706.78 \times 21.949 = 3{,}154{,}179 \text{ kg-cm}$$

Moment from compression steel reinforcement (M_{Cs})

$$C'_s = 8{,}375.1 \text{ kg}$$

$$\text{Arm} = \frac{h}{2} - d' = 30.00 - 4.60 = 25.40 \text{ cm}$$

$$M_{Cs} = 8{,}375.1 \times 25.40 = 212{,}728 \text{ kg-cm}$$

Moment from tension steel reinforcement (M_{Ts})

$$T_s = 157{,}080.0 \text{ kg}$$

$$\text{Arm} = d - \frac{h}{2} = 52.25 - 30.00 = 22.25 \text{ cm}$$

$$M_{Ts} = 157{,}080.0 \times 22.25 = 3{,}495{,}030 \text{ kg-cm}$$

Total nominal flexural strength (M_n) and design flexural strength (ϕM_n)

$$M_n = M_{Cc} + M_{Cs} + M_{Ts}$$

$$M_n = 3{,}154{,}179 + 212{,}728 + 3{,}495{,}030 = 6{,}861{,}937 \text{ kg-cm} = 68.619 \text{ t-m}$$

$$\phi M_n = 0.90 \times 68.619 = 61.757 \text{ t-m}$$

Design moment capacity (ϕM_n)

ϕM_n (61.757 t-m) ≥ $|M_u|$ (54 t-m) PASS

2.3. Shear capacity analysis

Concrete shear resistance from ACI 318-14 Table 22.5.7.1 (Axial Tension)

$$V_c = 0.53 \cdot \left(1 + \frac{P_u}{35 \cdot A_g}\right) \cdot \sqrt{f'_c} \cdot b \cdot d$$

$$\left(1 + \frac{P_u}{35 \cdot A_g}\right) = 1 + \frac{-5000}{35 \times 1800} = 1 - 0.07937 = 0.92063$$

$$V_c = 0.53 \times 0.92063 \times \sqrt{350} \times 1567.50 = 14{,}308 \text{ kg} = 14.31 \text{ tons}$$

$$\phi V_c = 0.75 \times 14.31 = 10.73 \text{ tons}$$

Provided tranverse stirrup shear strength (V_s - 2 Legs DB10@ 7 cm)

$$A_v = 2 \times \frac{\pi(1.0)^2}{4} = 1.571 \text{ cm}^2$$

$$V_s = \frac{A_v \cdot f_{yt} \cdot d}{s_{\text{prov}}} = \frac{1.571 \times 4000 \times 52.25}{7} = 46{,}905.57 \text{ kg} = 46.91 \text{ tons}$$

$$\left(\frac{A_v}{s}\right)_{\text{min}} = \max\left(\frac{0.2\sqrt{f'_c} \cdot b}{f_{yt}}, \frac{3.5 \cdot b}{f_{yt}}\right) = 0.02806$$

Check with actual stirrup reinforcement (2-legged DB10 @ 7 cm, $A_v = 1.571$ cm²)

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.571}{7.00} = 0.2244 \text{ cm}^2/\text{cm}$$

$\left(\frac{A_v}{s}\right)_{\text{prov}}$ $(0.2244 \text{ cm}^2/\text{cm}) \geq \left(\frac{A_v}{s}\right)_{\text{min}}$ $(0.02806 \text{ cm}^2/\text{cm})$ PASS

Check concrete crushing limit ($V_{s,\max}$)

$$V_{s,\max} = 2.1\sqrt{f'_c} \cdot b \cdot d = 2.1 \times \sqrt{350} \times 30.00 \times 52.25 = 61{,}583 \text{ kg} = 61.58 \text{ tons}$$

$V_{s,\max}$ (61.58 tons) ≥ V_s (46.91 tons) PASS

Check maximum stirrup spacing ($s_{\max}$) Table 9.7.6.2.2

$$\text{Threshold Limit} = 1.10\sqrt{f'_c} \cdot b \cdot d = 1.10 \times \sqrt{350} \times 30.00 \times 52.25 = 32{,}258 \text{ kg}$$

$$V_s \text{ (46.91 tons)} \geq \text{Threshold limit (32.26 tons)}$$

$$s_{\max} = \min\left(\frac{d}{4},\ 30.0\text{ cm}\right) = \min\left(\frac{52.25}{4},\ 30.0\right) = 13.06\text{ cm}$$

$s_{\max}$ (13.06 cm) ≥ s_{prov} (7.00 cm) PASS

Total design shear strength (ϕV_n)

$$\phi V_n = \phi_{\text{shear}} \cdot (V_c + V_s) = 0.75 \times (14.31 + 46.91) = 45.92\text{ tons}$$

ϕV_n (45.92 tons) ≥ V_u (30 tons) PASS

2.4. Torsion assessment

Threshold torsion calculation (T_{th})

$$A_{cp} = b \cdot h = 30.0 \times 60.0 = 1{,}800.00\text{ cm}^2, \quad p_{cp} = 2(b + h) = 180.0\text{ cm}$$

$$T_{th} = \phi_{\text{torsion}} \cdot 0.27\lambda\sqrt{f'_c} \cdot \left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1 + \frac{P_u}{0.33A_g\sqrt{f'_c}}}$$

$$T_{th} = 0.75 \times 0.27 \times 1.0 \times \sqrt{350} \times \left(\frac{1800^2}{180}\right) \times \sqrt{1 + \frac{-5000}{0.33 \times 1800 \times \sqrt{350}}} = 50{,}473\text{ kg-cm}$$

Torsion Design

T_{th} (0.505 t-m) ≥ T_u (0.300 t-m) NEGLECT TORSION

3. Critical section MID (span positive moment)

3.1 Geometry and factored demand

M_u = 11,000 kg-m = 11 t-m

V_u = 5,000 kg = 5 tons

T_u = 0 kg-m = 0 t-m

P_u = 12,000 kg = 12 tons

Top 2-DB12 (A'_s = 2 x 1.131 = 2.262 cm^2)

Bottom 2-DB20 (A_s = 2 x 3.142 = 6.283 cm^2)

Beam Applicability Limit Check (ACI 318-14 Section 9.3.3.1)

The factored axial compressive force on a flexural member shall not exceed ($0.1f'_cA_g$)

$$P_{\text{limit}} = 0.10f'_cA_g = 0.10 \times 350 \times 1{,}800 = 63{,}000\text{ kg} = 63\text{ tons}$$

P_{limit} (63 T) ≥ P_u (12 T) PASS (flexural member member provisions apply)

Effective depth (d) of tension steel (bottom)

$$y_1 = c_{\text{cov}} + d_{st} + \frac{d_b}{2} = 3.00 + 1.00 + \frac{2.00}{2} = 5.00 \text{ cm}$$

$$d_1 = h - y_1 = 60.00 - 5.00 = 55.00 \text{ cm}$$

Compression steel depth from top face (d')

$$d' = c_{\text{cov}} + d_{st} + \frac{d_{b,12}}{2} = 3.00 + 1.00 + \frac{1.20}{2} = 4.60 \text{ cm}$$

Rebar clearance and spacing limits (ACI 25.2.1)

$$s_{\text{clear}} = \frac{b - 2c_{\text{cov}} - 2d_{st} - n \cdot d_b}{n - 1} = \frac{30 - (2 \times 3.0) - (2 \times 1.0) - (2 \times 2.0)}{2 - 1} = 18.0 \text{ cm}$$

$$s_{\text{clear,min}} = \max(d_b,\ 2.5 \text{ cm}) = \max(2.0,\ 2.5) = 2.50 \text{ cm}$$

$s_{\text{clear}}(18.0 \text{ cm}) \geq s_{\text{clear,min}}(2.50 \text{ cm})$ PASS

3.2. Flexural strength analysis

Force equilibrium and neutral axis depth

$$C_c + C_s' - T_s = P_u$$

$$C_c + C_s' = P_u + T_s$$

Check yielding of tension steel T_s

Assume all layers of tension steel yield ($f_s = f_y = 4{,}000$)

$$T_s = A_s \cdot f_y = 6.283 \times 4000 = 25{,}132 \text{ kg}$$

Check yielding of compression steel (C'_s)

The stress in compression steel is $f'_s = \min(f_y,\ E_s\epsilon'_s)$ accounting for the displaced concrete area ($0.85f'_c = 297.5$ ksc)

Since neutral axis c is near d', compression steel remains elastic ($f'_s < f_y$).

$$f'_s = E_s \cdot \epsilon'_s = 2.04 \times 10^6 \times 0.003 \cdot \frac{c - d'}{c} = 6{,}120 \cdot \frac{c - 4.6}{c}$$

$$C'_s = A'_s(f'_s - 0.85f'_c) = 2.262\left(6{,}120 \cdot \frac{c - 4.6}{c} - 297.5\right) = 13{,}843.44 \cdot \frac{c - 4.6}{c} - 672.95$$

$$C_c = 0.85f'_c\beta_1 cb = 0.85 \times 350 \times 0.80 \times c \times 30.0 = 7{,}140.0 \cdot c$$

Solving equilibrium quadratic equation for neutral axis c:

$$7{,}140.0 \cdot c + 13{,}843.44\left(\frac{c - 4.60}{c}\right) - 672.95 = 25{,}132.0 + 12{,}000.0$$

$$7140.0 \cdot c^2 - 23{,}961.51 \cdot c - 63{,}679.82 = 0$$

$$c = 5.104 \text{ cm}$$

$$\epsilon'_s = 0.003 \cdot \left(\frac{c - d'}{c}\right) = 0.003 \cdot \left(\frac{5.104 - 4.60}{5.104}\right) = 0.000296 < \epsilon_y = 0.00196$$

(NOT YIELD CONFIRMED)

$$f'_s = E_s \times \epsilon'_s = 2.04 \times 10^6 \times 0.000296 = 604.3\ ksc < f_y = 4{,}000 \text{ ksc}$$

$$C'_s = A'_s \cdot (f'_s - 0.85 f'_c) = 2.262 \times (604.3 - 297.5) = 694.0 \text{ kg}$$

$$C_c = 0.85 f'_c \beta_1 cb = 0.85 \times 350 \times 0.80 \times 5.104 \times 30.0 = 36{,}442.6 \text{ kg}$$

Check the depth of the compression block (a) compared to (d')

$$a = \beta_1 \cdot c = 0.80 \times 5.104 = 4.083 \text{ cm}$$

$$a\ (4.083 \text{ cm}) < d'\ (4.60 \text{ cm})$$

Tension Strain Check (ϵ_t) and Strength Reduction Factor (φ)

$$\epsilon_{t,\text{outer}} = 0.003 \cdot \left(\frac{d - c}{c}\right) = 0.003 \cdot \left(\frac{55.00 - 5.104}{5.104}\right) = 0.02933$$

$$\epsilon_{t,\text{outer}} = 0.02933 \geq 0.00500$$

$$\phi = 0.900 \text{ (Tension} - \text{Controlled)}$$

$$A_{s,\min} = \max\left(\frac{14}{f_y}, \frac{0.8\sqrt{f'_c}}{f_y}\right) \cdot b \cdot d = \max\left(\frac{14}{4000}, \frac{0.8\sqrt{350}}{4000}\right) \cdot 30 \cdot 55.00 = 6.174 \text{ cm}^2$$

Check with actual reinforcement (2-DB20 = 6.283 cm^2):

$$A_{s,prov}(6.283\ cm^2) \geq A_{s,min}(6.174\ cm^2) \text{ PASS}$$

Moment from concrete compression (M_{Cc})

$$C_c = 0.85 f'_c \beta_1 cb = 0.85 \times 350 \times 0.80 \times 5.104 \times 30.0 = 36{,}442.56 \text{ kg}$$

$$\text{Arm} = \frac{h}{2} - \frac{a}{2} = 30.00 - \frac{4.083}{2} = 27.959 \text{ cm}$$

$$M_{Cc} = 36{,}442.6 \times 27.959 = 1{,}018{,}899 \text{ kg-cm}$$

Moment from compression steel reinforcement (M_{Cs})

$$C'_s = 694.0 \text{ kg}$$

$$\text{Arm} = \frac{h}{2} - d' = 30.00 - 4.60 = 25.40 \text{ cm}$$

$$M_{Cs} = 694.0 \times 25.40 = 17{,}628.00 \text{ kg-cm}$$

Moment from tension steel reinforcement (M_{Ts})

$$T_s = 25{,}132.0 \text{ kg}$$

$$\text{Arm} = d - \frac{h}{2} = 55.00 - 30.00 = 25.00 \text{ cm}$$

$$M_{Ts} = 25{,}132.0 \times 25.00 = 628{,}300 \text{ kg-cm}$$

Total nominal flexural strength (M_n) and design flexural strength (ϕM_n)

$$M_n = M_{Cc} + M_{Cs} + M_{Ts}$$

$$M_n = 1{,}018{,}899 + 17{,}628 + 628{,}300 = 1{,}664{,}827 \text{ kg-cm} = 16.648 \text{ t-m}$$

$$\phi M_n = 0.90 \times 16.648 = 14.983 \text{ t-m}$$

Design moment capacity (ϕM_n)

ϕM_n (14.983 t-m) ≥ $|M_u|$ (11 t-m) PASS

3.3. Shear capacity analysis

Concrete shear resistance from ACI 318-14 Table 22.5.6.1 (axial compression)

$$V_c = 0.53 \cdot \left(1 + \frac{P_u}{140 \cdot A_g}\right) \cdot \sqrt{f'_c} \cdot b \cdot d$$

$$\left(1 + \frac{P_u}{140 \cdot A_g}\right) = 1 + \frac{12{,}000}{140 \times 1800} = 1 + 0.047619 = 1.047619$$

$$V_c = 0.53 \times 1.047619 \times 18.7083 \times 1650.00 = 17{,}139.48 \text{ kg} = 17.14 \text{ tons}$$

$$\phi V_c = 0.75 \times 17.14 = 12.86 \text{ tons}$$

Provided tranverse stirrup shear strength (V_s - 2 Legs DB10@ 7 cm)

$$A_v = 2 \times \frac{\pi(1.00)^2}{4} = 1.571 \text{ cm}^2$$

$$V_s = \frac{A_v \cdot f_{yt} \cdot d}{s_{\text{prov}}} = \frac{1.571 \times 4000 \times 55.00}{7} = 49{,}374.29 \text{ kg} = 49.374 \text{ tons}$$

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.571}{7.0} = 0.224 \text{ cm}^2/\text{cm}$$

$$\left(\frac{A_v}{s}\right)_{\min} = \max\left(\frac{0.2\sqrt{f'_c} \cdot b}{f_{yt}}, \frac{3.5 \cdot b}{f_{yt}}\right) = 0.02806$$

$\left(\frac{A_v}{s}\right)_{\text{prov}}$ (0.224 cm²/cm) ≥ $\left(\frac{A_v}{s}\right)_{\min}$ (0.02806 cm²/cm) PASS

Check concrete crushing limit ($V_{s,\max}$)

$$V_{s,\max} = 2.1\sqrt{f'_c} \cdot b \cdot d = 2.1 \times \sqrt{350} \times 30.00 \times 55.00 = 64{,}824 \text{ kg} = 64.82 \text{ tons}$$

$V_{s,\max}$ (64.82 tons) $\geq V_s$ (49.37 tons) PASS

Check maximum stirrup spacing ($s_{\max}$) Table 9.7.6.2.2

$$\text{Threshold Limit} = 1.10\sqrt{f'_c} \cdot b \cdot d = 1.10 \times \sqrt{350} \times 30.00 \times 55.00 = 33.96 \text{ tons}$$

V_s (49.37 tons) > Threshold limit (33.96 tons)

$$s_{\max} = \min\left(\frac{d}{4}, 30.0 \text{ cm}\right) = \min\left(\frac{55.00}{4}, 30.0\right) = 13.75 \text{ cm}$$

$s_{\max}$ (7.00 cm) $\leq s_{\text{prov}}$ (13.75 cm) PASS

Total design shear strength (ϕV_n)

$$\phi V_n = \phi_{\text{shear}} \cdot (V_c + V_s) = 0.75 \times (17.14 + 49.37) = 49.88 \text{ tons}$$

ϕV_n (49.88 tons) $\geq V_u$ (5.0 tons) PASS

3.4. Torsion assessment

Threshold torsion calculation (T_{th})

Threshold torsion calculation (T_{th}) (ACI 318-14 Sec 22.7.4.1)

$$A_{cp} = b \cdot h = 30.00 \times 60.00 = 1{,}800.0 \text{ cm}^2, \quad p_{cp} = 2(b + h) = 180.0 \text{ cm}$$

$$T_{th} = \phi_{\text{torsion}} \cdot 0.27\lambda\sqrt{f'_c} \cdot \left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1 + \frac{P_u}{0.33A_g\sqrt{f'_c}}}$$

$$T_{th} = 0.75 \times 0.27 \times 1.0 \times \sqrt{350} \times \left(\frac{1800^2}{180}\right) \times \sqrt{1 + \frac{12{,}000}{0.33 \times 1800 \times \sqrt{350}}} = 98{,}140 \text{ kg-cm}$$

Torsion Design

T_{th} (0.98 t-m) $\geq T_u$ (0.0 t-m) NEGLECT TORSION

4. Critical section END (support negative moment)

4.1. Geometry and factored demand

M_u = -22,000 kg-m = -22 t-m

V_u = 45,000 kg = 45 tons

T_u = 1,500 kg-m = 1.5 t-m

P_u = 6,000 kg = 6 tons

Top 3-DB25 (A_s = 3 x 4.909 = 14.727 cm^2)

Bottom 3-DB16 (A'_s = 3 x 2.011 = 6.033 cm^2)

Side Steel (Web Reinforcement) 4-DB16 ($A_{l,\text{side}} = 4 \times 2.011 = 8.044\ \text{cm}^2$)

Beam Applicability Limit Check (ACI 318-14 Section 9.3.3.1)

The factored axial compressive force on a flexural member shall not exceed ($0.1f'_cA_g$)

$$P_{\text{limit}} = 0.10f'_cA_g = 0.10 \times 350 \times 1{,}800 = 63{,}000\ \text{kg} = 63.0\ \text{tons}$$

$P_{limit}(63.0\ \text{tons}) \geq P_u(6.0\ \text{tons})$ PASS (flexural member provisions apply)

Effective depth (d) of tension steel (top)

$$y_1 = c_{\text{cov}} + d_{st} + \frac{d_b}{2} = 3.00 + 1.00 + \frac{2.50}{2} = 5.25\ \text{cm}$$

$$d_1 = h - y_1 = 60.00 - 5.250 = 54.750\ \text{cm}$$

Compression steel depth from bottom face (d_2)

$$d_2 = c_{\text{cov}} + d_{st} + \frac{d_{b,12}}{2} = 3.00 + 1.00 + \frac{1.60}{2} = 4.80\ \text{cm}$$

4.2. Flexural strength analysis

Force equilibrium and neutral axis depth

$$C_c + C'_s - T_s = P_u$$

$$C_c + C'_s = P_u + T_s$$

Check yielding of tension steel T_s

Assume tension steel yields ($f_s = f_y$ = 4,000 ksc)

$$T_s = A_s \cdot f_y = 14.727 \times 4{,}000 = 58{,}908.0\ \text{kg}$$

$$T_s + P_u = 58{,}908.0 + 6{,}000.0 = 64{,}908.0\ \text{kg}$$

Check yielding of compression steel (C'_s)

The stress in compression steel is $f'_s = \min(f_y,\ E_s\epsilon'_s)$ accounting for the displaced concrete area ($0.85f'_c = 297.5$ ksc)

$$f'_s = E_s \cdot \epsilon'_s = 2.04 \times 10^6 \times 0.003 \cdot \frac{c - d'}{c} = 6{,}120 \cdot \frac{c - 4.8}{c}$$

$$C'_s = A'_s(f'_s - 0.85f'_c) = 6.033\left(6{,}120 \cdot \frac{c - 4.8}{c} - 297.5\right) = 36{,}921.96 \cdot \frac{c - 4.8}{c} - 1{,}794.82$$

$$C_c = 0.85f'_c\beta_1 cb = 0.85 \times 350 \times 0.80 \times c \times 30.0 = 7{,}140.0 \cdot c$$

Solving equilibrium quadratic equation for neutral axis c:

$$7{,}140.0 \cdot c + 36{,}921.96\left(\frac{c - 4.80}{c}\right) - 1{,}794.82 = 64{,}908.0$$

$$7{,}140.0 \cdot c^2 - 29{,}780.86 \cdot c - 177{,}225.41 = 0$$

$$c = 7.483 \text{ cm}$$

$$\epsilon'_s = 0.003 \cdot \left(\frac{7.483 - 4.80}{7.483}\right) = 0.001076 < \epsilon_y(0.00196) \quad \text{(NOT YIELD)}$$

$$f'_s = 2.04 \times 10^6 \times 0.001076 = 2{,}194.50 \text{ ksc} < f_y(4{,}000 \text{ ksc})$$

$$C'_s = A'_s \cdot (f'_s - 0.85f'_c) = 6.033 \times (2{,}194.50 - 297.50) = 11{,}444.60 \text{ kg}$$

$$C_c = 0.85f'_c\beta_1 cb = 0.85 \times 350 \times 0.80 \times 7.483 \times 30.0 = 53{,}428.62 \text{ kg}$$

Check the depth of the compression block (a) compared to (d')

$$a = \beta_1 \cdot c = 0.80 \times 7.483 = 5.986 \text{ cm} > \text{d' (4.80 cm)}$$

Tension Strain Check (ϵ_t) and Strength Reduction Factor (φ)

$$\epsilon_t = 0.003 \cdot \left(\frac{d - c}{c}\right) = 0.003 \cdot \left(\frac{54.75 - 7.483}{7.483}\right) = 0.01895$$

$$\epsilon_{t,\text{outer}} = 0.01895 \geq 0.00500$$

$$\phi = 0.900 \text{ (Tension} - \text{Controlled)}$$

$$A_{s,\min} = \max\left(\frac{14}{f_y}, \frac{0.8\sqrt{f'_c}}{f_y}\right) \cdot b \cdot d = \max\left(\frac{14}{4000}, \frac{0.8\sqrt{350}}{4000}\right) \cdot 30 \cdot 54.75 = 6.148 \text{ cm}^2$$

$A_{s,prov}(14.727\ cm^2) \geq A_{s,min}(6.148\ cm^2)$ PASS

Moment from concrete compression (M_{Cc})

$$\text{Arm} = \frac{h}{2} - \frac{a}{2} = 30.00 - \frac{5.986}{2} = 27.007 \text{ cm}$$

$$M_{Cc} = 53{,}428.62 \times 27.007 = 1{,}442{,}946.74 \text{ kg-cm}$$

Moment from compression steel reinforcement (M_{Cs})

$$\text{Arm} = \frac{h}{2} - d' = 30.00 - 4.8 = 25.20 \text{ cm}$$

$$M_{Cs} = 11{,}444.60 \times 25.20 = 288{,}403.92 \text{ kg-cm}$$

Moment from tension steel reinforcement (M_{Ts} about the centroid axis h/2)

$$\text{Arm} = d - \frac{h}{2} = 54.75 - 30.00 = 24.75 \text{ cm}$$

$$M_{Ts} = 58{,}908.00 \times 24.75 = 1{,}457{,}973 \text{ kg-cm}$$

Total nominal flexural strength (M_n) and design flexural strength (ϕM_n)

$$M_n = M_{Cc} + M_{Cs} + M_{Ts}$$

$$M_n = 1{,}442{,}946.74 + 288{,}403.92 + 1{,}457{,}973 = 3{,}189{,}323.66 \text{ kg-cm} = 31.893 \text{ t-m}$$

$$\phi M_n = 0.90 \times 31.893 = 28.704 \text{ t-m}$$

Design moment capacity (ϕM_n)

ϕM_n (28.704 t-m) ≥ $|M_u|$ (22 t-m) PASS

4.3. Shear capacity analysis

Concrete shear resistance from ACI 318-14 Table 22.5.6.1 (Axial Compression)

$$V_c = 0.53 \cdot \left(1 + \frac{P_u}{140 \cdot A_g}\right) \cdot \sqrt{f'_c} \cdot b \cdot d$$

$$\left(1 + \frac{P_u}{140 \cdot A_g}\right) = 1 + \frac{6{,}000}{140 \times 1800} = 1 + 0.02381 = 1.02381$$

$$V_c = 0.53 \times 1.02381 \times \sqrt{350} \times 30 \times 54.75 = 16{,}674 \text{ kg} = 16.67 \text{ tons}$$

Provided Traverse stirrup shear strength (V_s - 2 Legs DB10@ 7 cm)

$$A_v = 2 \times \frac{\pi(1.0)^2}{4} = 1.571 \text{ cm}^2$$

$$V_s = \frac{A_v \cdot f_{yt} \cdot d}{s_{\text{prov}}} = \frac{1.571 \times 4000 \times 54.75}{7} = 49{,}150 \text{ kg} = 49.15 \text{ tons}$$

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.571}{7.0} = 0.2244 \text{ cm}^2/\text{cm}$$

$$\left(\frac{A_v}{s}\right)_{\min} = \max\left(\frac{0.2\sqrt{f'_c} \cdot b}{f_{yt}}, \frac{3.5 \cdot b}{f_{yt}}\right) = 0.0281 \text{ cm}^2/\text{cm}$$

$\left(\frac{A_v}{s}\right)_{\text{prov}}$ (0.2244 cm²/cm) ≥ $\left(\frac{A_v}{s}\right)_{\min}$ (0.0281 cm²/cm) PASS

Total design shear strength (ϕV_n)

$$\phi V_n = \phi_{\text{shear}} \cdot (V_c + V_s) = 0.75 \times (16.67 + 49.15) = 49.37 \text{ tons}$$

ϕV_n (49.37 tons) ≥ V_u (47.15 tons) PASS

Check concrete crushing limit ($V_{s,\max}$)

$$V_{s,\text{req}} = \frac{V_u}{\phi} - V_c = \frac{45{,}000}{0.75} - 16{,}674 = 43{,}326 \text{ kg} = 43.33 \text{ tons}$$

$$V_{s,\max} = 2.1\sqrt{f'_c} \cdot b \cdot d = 2.1 \times \sqrt{350} \times 30.0 \times 54.75 = 64{,}530 \text{ kg} = 64.53 \text{ tons}$$

$V_{s,\max}$ (64.53 tons) ≥ V_s (43.33 tons) PASS

Check maximum stirrup spacing ($s_{\max}$) Table 9.7.6.2.2

$$\text{Threshold Limit} = 1.10\sqrt{f'_c} \cdot b \cdot d = 1.10 \times \sqrt{350} \times 30.0 \times 54.75 = 33.80 \text{ tons}$$

$$V_s \text{ (43.33 tons)} > \text{Threshold limit (33.80 tons)}$$

$$s_{\max} = \min\left(\frac{d}{4}, 30.0 \text{ cm}\right) = \min\left(\frac{54.75}{4}, 30.0\right) = 13.69 \text{ cm}$$

$s_{\max}$ (13.69 cm) ≥ s_{prov} (7.00 cm) PASS

4.4. Torsion assessment

$$A_{cp} = 30.00 \times 60.00 = 1{,}800.00 \text{ cm}^2,\ p_{cp} = 2(30.00 + 60.00) = 180.00 \text{ cm}$$

Threshold torsion calculation (T_{th}) accounting for axial compression (ACI 318-14 Sec 22.7.4.1)

$$T_{th} = \phi_{\text{torsion}} \cdot 0.27\lambda\sqrt{f'_c} \cdot \left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1 + \frac{P_u}{0.33A_g\sqrt{f'_c}}}$$

$$T_{th} = 0.75 \times 0.27 \times 1.0 \times \sqrt{350} \times \left(\frac{1800^2}{180}\right) \times \sqrt{1 + \frac{6{,}000}{0.33 \times 1800 \times \sqrt{350}}} = 84{,}446 \text{ kg-cm}$$

$$T_{th} = 0.845 \text{ t-m}$$

Torsion Design

T_{th} (0.845 t-m) ≤ T_u (1.500 t-m) REQUIRE TORSION DESIGN

Torsion Critical

$$T_{cr} = \phi_{\text{torsion}} \cdot 1.10\sqrt{f'_c} \cdot \left(\frac{A_{cp}^2}{p_{cp}}\right) = 0.75 \times 1.10 \times \sqrt{350} \times \left(\frac{1{,}800.00^2}{180.00}\right) = 277{,}818 \text{ kg-cm}$$

$$T_{cr} = 2.778 \text{ t-m}$$

T_{cr} (2.778 t-m) ≥ T_u (1.500 t-m)*PASS*

Core Geometry Properties

Stirrup using DB10: d_{st} = 1.0 cm (A_v = 1.571 cm^2 for 2 legs), f_{yt} = 4,000 ksc

$$x_1 = b - 2\left(c_{cov} + \frac{d_v}{2}\right) = 30.0 - 2(3.0 + 0.5) = 23.0 \text{ cm}$$

$$y_1 = h - 2\left(c_{cov} + \frac{d_v}{2}\right) = 60.0 - 2(3.0 + 0.5) = 53.0 \text{ cm}$$

Area and perimeter enclosed by the centerline of the stirrup (A_{oh}, p_h, A_o)

$$A_{oh} = x_1 \cdot y_1 = 23.00 \times 53.00 = 1{,}219.00 \text{ cm}^2$$

$$p_h = 2(x_1 + y_1) = 2(23.00 + 53.00) = 152.00 \text{ cm}$$

$$A_o = 0.85 \cdot A_{oh} = 0.85 \times 1{,}219.00 = 1{,}036.15 \text{ cm}^2$$

Section dimension check to prevent web crushing under combined actions
Shear stress due to shear force (v_u)

$$v_u = \frac{V_u}{b \cdot d} = \frac{45{,}000}{30.0 \times 54.75} = 27.397 \text{ kg/cm}^2$$

Shear stress due to torsion (v_t)

$$v_t = \frac{T_u \cdot p_h}{1.7 A_{oh}^2} = \frac{150{,}000 \times 152.0}{1.7 \times (1{,}219.0)^2} = 9.026 \text{ kg/cm}^2$$

Combined shear stress

$$\sqrt{v_u^2 + v_t^2} = \sqrt{27.397^2 + 9.026^2} = 28.846 \text{ kg/cm}^2$$

Maximum capacity limit of the concrete section (v_{max} per ACI 318-14 Sec 22.7.7.1)

$$v_{\max} = \phi_{\text{shear}} \cdot \left(\frac{V_c}{b \cdot d} + 2.1\sqrt{f'_c} \right) = 0.75 \times \left(\frac{16{,}674}{30.0 \times 54.75} + 2.1\sqrt{350} \right) = 37.08 \text{ kg/cm}^2$$

$$v_{\max}(37.08 \text{ kg/cm}^2) \geq \sqrt{v_u^2 + v_t^2}\,(28.846 \text{ kg/cm}^2) \quad \text{PASS}$$

Combined Stirrup Design
Area of one outer stirrup leg for torsion (A_t/s at $\theta = 45°$)

$$\frac{A_t}{s} = \frac{T_u}{2\phi_{\text{tor}} A_o f_{yt} \cot\theta} = \frac{150{,}000}{2 \times 0.75 \times 1{,}036.15 \times 4{,}000 \times 1.0} = 0.02413 \text{ cm}^2/\text{cm}$$

Total required stirrup area ratio for shear and torsion (ACI 318-14 Sec 22.7.6.1) Shear stirrup demand with $\phi_{\text{shear}} = 0.75$

$$V_{s,\text{req}} = \frac{V_u}{\phi_{\text{shear}}} - V_c = \frac{45{,}000}{0.75} - 16{,}674 = 60{,}000 - 16{,}674 = 43{,}326 \text{ kg}$$

$$\left(\frac{A_v}{s}\right)_{\text{shear}} = \frac{V_{s,\text{req}}}{f_{yt} \cdot d} = \frac{43{,}326}{4{,}000 \times 54.75} = 0.19784 \text{ cm}^2/\text{cm}$$

Torsion stirrup demand for 2 outer legs

$$2\left(\frac{A_t}{s}\right) = 2 \times 0.02413 = 0.04826 \text{ cm}^2/\text{cm}$$

Total area ratio for 2-leg stirrups (A_v/s)

$$\left(\frac{A_v}{s}\right)_{\text{comb}} = \left(\frac{A_v}{s}\right)_{\text{shear}} + 2\left(\frac{A_t}{s}\right) = 0.19784 + 0.04826 = 0.24610 \text{ cm}^2/\text{cm}$$

Required spacing based on strength (using 2-legs DB10, Av = 1.571 cm²)

$$s_{\text{req}} = \frac{A_v}{(A_v/s)_{\text{comb}}} = \frac{1.571}{0.24610} = 6.38 \text{ cm}$$

Maximum allowable spacing check per standard (s_{allow})

$$s_{\text{max,tor}} = \min\left(\frac{p_h}{8}, 30.0 \text{ cm}\right) = \min\left(\frac{152.00}{8}, 30.0\right) = 19.00 \text{ cm}$$

$$s_{\text{max,geom}} = s_{\text{max,shear}} = 13.69 \text{ cm} \quad (\text{from } d/4 \text{ in 3.3})$$

$$s_{\text{allow}} = \min(s_{\text{req}}, s_{\text{max,tor}}, s_{\text{max,shear}}) = \min(6.38, 19.00, 13.69) = 6.38 \text{ cm}$$

Verification of provided stirrups (2-Legs DB10 @ 7.0 cm, $s_{\text{prov}} = 7.00$ cm)

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.571}{7.00} = 0.22443 \text{ cm}^2/\text{cm}$$

$\left(\frac{A_v}{s}\right)_{\text{prov}} (0.2244 \text{ cm}^2/\text{cm}) < \left(\frac{A_v}{s}\right)_{\text{comb,req}} (0.2461 \text{ cm}^2/\text{cm})$ NOTPASS (UNSAFE)

s_{allow} (6.38 cm) < s_{prov} (7.00 cm) NOTPASS (stirrup Area Insufficient)

The provided area of stirrups is greater than the required area.

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.571}{7.0} = 0.2244 \text{ cm}^2/\text{cm}$$

Remaining actual area of stirrups for torsion (2 legs) only

$$2\left(\frac{A_t}{s}\right)_{\text{spare}} = 0.2244 - 0.1978 = \frac{0.0266 \text{ cm}^2}{\text{cm}}$$

$$\frac{A_t}{s} = 0.0133 \text{ cm}^2/\text{cm}$$

Used to calculate the actual effective torsional strength of the structure:

$$\phi T_{n,t} = 0.75 \times (2 \times 1{,}036.15) \times 0.0133 \times 4{,}000 \times 1.0 = 82{,}685 \text{ kg-cm} = 0.83 \text{ t-m}$$

Spare stirrup capacity check under combined actions:

Remaining stirrup area for shear after resisting full torsion:

$$\left(\frac{A_v}{s}\right)_{\text{shear,rem}} = \left(\frac{A_v}{s}\right)_{\text{prov}} - 2\left(\frac{A_t}{s}\right) = 0.22443 - 0.04826 = 0.17617 \text{ cm}^2/\text{cm}$$

$$V_{s,\text{rem}} = 0.17617 \times 4{,}000 \times 54.75 = 38{,}581 \text{ kg} = 38.58 \text{ tons}$$

$$\phi V_n = \phi_{\text{shear}} \cdot \left(V_c + V_{s,\text{rem}}\right) = 0.75 \times (16.67 + 38.58) = 41.44 \text{ tons}$$

ϕV_n (41.44 tons) $< V_u$ (45.00 tons) NOTPASS (combined shear overstress)

Longitudinal Torsion Reinforcement Design (Al)

Required longitudinal reinforcement area due to torsion (Al)

$$A_l = \left(\frac{A_t}{s}\right) p_h \left(\frac{f_{yt}}{f_y}\right) \cot^2\theta = 0.02413 \times 152.00 \times \left(\frac{4{,}000}{4{,}000}\right) \times (1.0)^2 = 3.668 \text{ cm}^2 = 366.8 \text{ mm}^2$$

Minimum longitudinal reinforcement area (Al,min)

$$\left(\frac{A_t}{s}\right)_{\text{clamped}} = \max\left(A_t/s, \frac{1.78 \cdot b}{f_{yt}}\right) = \max\left(0.02413, \frac{1.78 \times 30.0}{4{,}000}\right) = 0.02413 \text{ cm}^2/\text{cm}$$

$$A_{l,min} = \frac{1.33\sqrt{f'_c} \cdot A_{cp}}{f_y} - \left(\frac{A_t}{s}\right)_{\text{clamped}} p_h \left(\frac{f_{yt}}{f_y}\right)$$

$$A_{l,min} = \frac{1.33 \times \sqrt{350} \times 1{,}800}{4{,}000} - (0.02413 \times 152.0 \times 1.0) = 7.538 \text{ cm}^2 = 753.8 \text{ mm}^2$$

Governing longitudinal reinforcement area ($A_{l,req}$)

$$A_{l,\text{req}} = \max\left(A_l, A_{l,\text{min}}\right) = 7.538 \text{ cm}^2 = 753.8 \text{ mm}^2$$

Check against provided side face reinforcement (Side Rebar 4-DB16)

$$A'_{s,\text{prov}} = 4 \times 2.011 = 8.044 \text{ cm}^2 = 804.4 \text{ mm}^2$$

$A'_{s,prov}$ (804.4 mm²) ≥ $A_{l,req}$ (753.8 mm²) PASS

Back-calculation from actual longitudinal bars 4-DB16 ($A'_{s,prov}$ = 804.4 mm²)

$$T_{n,l} = \frac{2A_o \cdot A_{l,\text{prov}} \cdot f_y}{p_h \cdot \cot\theta} = 0.75 \times \frac{2 \times 1{,}036.15 \times 8.044 \times 4{,}000}{152.0 \times 1.0} = 3.290 \text{ t-m}$$

Back-calculation from remaining spare stirrup capacity ($A_{t/s}$ = 0.0294 cm²/cm)

$$T_{n,t} = \frac{2A_o A_t f_{yt}}{s} \cot\theta$$

$$\phi T_{n,t} = 0.75 \times (2 \times 1{,}036.15) \times \left(\frac{45{,}090}{1{,}533{,}000}\right) \times 4{,}000 \times 1.0 = 1.828\ t-m$$

$$\phi T_n = \min\left(\phi T_{n,l}, \phi T_{n,t}\right) = \min(3.290 \text{ t-m}, 1.829 \text{ t-m}) = 1.829 \text{ t-m}$$

Section	Governing Action Demand	Design Capacity	D/C Ratio (UR)	ACI 318-14 Compliance Status

INITIAL *(Support Negative)*	$M_u = -54.0$ t-m	$\phi M_n = 61.76$ t-m	0.87	PASS (Tension-Controlled at d_t, Doubly Reinf., Yielded Comp. Steel)
	$V_u = 30.0$ tons	$\phi V_n = 45.92$ tons	0.65	PASS (Stirrups 2-Legs DB10@70mm Adequate, $s \leq s_{\max} =$ 13.06 cm)
	$T_u = 0.30$ t-m	$T_{\text{th}} = 0.51$ t-m	0.00	PASS (Neglect Torsion per $T_u \leq T_{\text{th}}$ accounting for axial tension)
MID *(Span Positive)*	$P_u = 12.0$ tons	$P_{\text{limit}} = 63.0$ tons	0.19	PASS (Axial Compression satisfies Beam Limit $P_u \leq 0.10 f'_c A_g$)
	$M_u = +11.0$ t-m	$\phi M_n = 14.98$ t-m	0.73	PASS (Tension-Controlled, Doubly Reinf., Elastic Comp. Steel)
	$V_u = 5.0$ tons	$\phi V_n = 49.88$ tons	0.10	PASS ($V_u \leq 0.5\phi V_c$, Stirrups DB10@70mm Provided)
	$T_u = 0$ t-m	$T_{\text{th}} = 0.98$ t-m	0.00	PASS (Neglect Torsion per $T_u \leq T_{\text{th}}$ accounting for axial compression)
END *(Support Negative)*	$M_u = -22.0$ t-m	$\phi M_n = 28.70$ t-m	0.77	PASS (Tension-Controlled, Doubly Reinf., Elastic Comp. Steel)
	$V_u = 45.0$ tons	$\phi V_n = 49.37$ tons	0.91	PASS (Stirrups 2-Legs DB10@70mm Adequate, $s \leq s_{\max} =$ 13.69 cm)
	$T_u = 1.50$ t-m	$\phi T_n = 0.83$ t-m	1.10	**FAIL** (UNSAFE: Combined stirrup ratio $(A_v/s)_{\text{req}} =$ 0.246 cm^2/cm exceeds provided 0.224 $\text{cm}^2/$ cm; deficient by 9.7%)